

Brady (Bowe) Tian

tian1@umd.edu • Website • LinkedIn • College Park, MD, USA

PROFILE

Experienced in AI research and development, focusing on large language models, interpretability (causal, representation learning), and computer vision. Recognized through publications and hands-on research, skilled in **Python, C++, SQL, and HTML/CSS/JavaScript**. Focused on Representation Engineering in LLMs and trustworthy AI, demonstrating a commitment to building robust and interpretable AI systems.

EDUCATION

◆ University of Maryland, College Park, MD, USA

Second-year Ph.D. student of Electrical and Computer Engineering, expected in May 2029

◆ Wuhan University, Wuhan, CHN

Bachelor's degree in Engineering in Information Security, received in June 2024

Cumulative GPA: 3.9/4

EXPERIENCE

◆ Research Assistant advised by Dr. [Ang Li](#) (06/2024 - present)

Propose a structured theory on how deep neural networks form **human-interpretable concepts**. It introduces the **Input-Space Linearity Hypothesis (ISLH)**, which posits that abstract concepts like "honesty" do not emerge mysteriously within the network, but are instead amplified from specific, concept-aligned directions already present in the initial input data. Through empirical validation on a **Vision-Language Model (VLM)**, the research shows that these conceptual representations become increasingly stable and consistent as they propagate through the network's layers, ultimately providing a clearer foundation for building more transparent, robust, and controllable AI.

As of **trustworthy AI**, developed EXogenous Causal reasoning (EXOC), a novel causal inference framework that utilizes auxiliary variables to enhance counterfactual fairness in general ML models. Used counterfactual scenarios to ensure an enhanced accuracy and fairness balance. The paper "**Towards counterfactual fairness through auxiliary variables**" has been accepted to ICLR 2025.

◆ Research Intern advised by Dr. [Zhewei Yao](#) (06/2025 - 08/2025, 09/2025 - present)

Summer internship at Snowflake, designed and implemented a context-aware **TextGrad auto-prompting** framework, treating prompts as learnable parameters optimized with gradient-based updates informed by schema structure and execution feedback. Built a customized pipeline including data augmentation, lightweight fine-tuning, enabling prompts to dynamically adapt to new databases. Applied **LoRA fine-tuning** on top of the auto-prompting system to further refine model parameters in a resource-efficient manner. Conducted extensive experiments on the **BIRD benchmark** across 11 heterogeneous databases, achieving up to +2% execution accuracy improvement over SOTA Text2SQL pipeline.

◆ Research Assistant advised by Dr. [Yanning Shen](#), Fairness on Vision Transformers (06/2023 - 10/2023)

Leveraged **adaptive masking** to enhance the fairness-accuracy tradeoff in **Vision Transformers (ViTs)**, a **deep neural network in computer vision**. Developed backward-propagation hooked masking in the **attention mechanism** and **contrastive learning** based distance loss, achieved a 6.72% improvement in accuracy compared to leading alternatives with similar fairness. Furthermore, interpretability, efficiency, and computation overhead are carefully crafted and analyzed. The paper "FairViT: Fair Vision Transformer via Adaptive Masking" has been accepted in the European Conference on Computer Vision (ECCV 2024).

◆ Research Assistant advised by Dr. [Chuang Gan](#), Rapper Pose Recognition and Generation (09/2023 - 11/2023)

Collaborated with Prof. Chuang Gan and Mr. Jiaben Chen to refactor and modularize large-scale codebases for OpenPose and TALKSHOW, improving code maintainability and motion detection. Integrated the **YOLOv3 object detection algorithm** to localize human subjects in YouTube videos, and developed a **motion preprocessing pipeline** that extracts, aligns, and normalizes pose sequences. Strengthened expertise in computer vision, multi-person pose estimation, and data-driven action synthesis.

◆ Research Assistant advised by Dr. [Qian Wang](#), Backdoor on Transformers (10/2022 - 06/2023)

Developed a trigger scope limitation strategy to enhance the stealthiness of **backdoor attacks in transformers** and manipulated the attention mechanism through "Attention Diffusion" to improve attack flexibility. Implemented these methods, achieving over a 25% improvement in stealthiness and efficiency compared to baseline models. Strengthened expertise in **trustworthy learning** and **foundations of LLMs**. The paper "An Effective and Resilient Backdoor Attack Framework against Deep Neural Networks and Vision Transformers" has been accepted in IEEE Transactions on Dependable and Secure Computing (TDSC 2025).

Extended the proposed **Quality of Experience (QoE)** attack method for both DNN and ViTs, achieving an attack success rate up to 82% higher than baseline methods at low poison ratios and maintaining high QoE in backdoored samples.

◆ Research Assistant co-advised by Dr. [Meng Xue](#) and Dr. [Xueluan Gong](#), Dry Eye Disease Detection (01/2023 - 05/2023)

Involved skills in **medical AI application**. Participated in radar-based detection for dry eye disease, offering a convenient and contactless screening method. Analyzed a focal loss-based **Transformer** model in Colab, conducted extensive ablation studies, and reorganized code to implement essential functions like data enhancement, dataset splitting, and model fine-tuning. These efforts culminated in the acceptance of the paper, "SDE: Early Screening for Dry Eye Disease with Wireless Signals" in Ubicomp/IMWUT 2024.

SELECTED PUBLICATIONS

B. Tian, Z. Wang, S. He, W. Ye, G. Sun, Y. Dai, Y. Wu, A. Li. [Towards counterfactual fairness through auxiliary variables](#). (ICLR 2025)

B. Tian, R. Du, Y. Shen. [FairViT: Fair Vision Transformer via Adaptive Masking](#). (ECCV 2024)

X. Gong*, **B. Tian***, M. Xue, Y. Wu, Y. Chen, Q. Wang. [An Effective and Resilient Backdoor Attack Framework against Deep Neural Networks and Vision Transformers](#). (TDSC 2025)

HONORS

Lei Jun Scholarship (2024)

Outstanding Student (in 2021, 2022 and 2023)

SKILLS

• Languages: C/C++, Python, SQL, HTML/CSS/JavaScript, MATLAB, Markdown

• Tools: SPSS, VS Code, Jupyter, LATEX, Github, Pytorch, Tensorflow, Conda, Docker

• Research interests: Fairness, Causal Reasoning, AI Security (Backdoor, Data Poisoning), Computer Vision, Vision Transformers, Deep Neural Networks, Natural Language Processing, Large Language Models, AI4Science, Multimodal Models, Federated Learning, Generative AI, Model Inversion, Adversarial Training, Interpretability, Representation Learning